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Roll No : 89 **Batch :** T21

Experiment No. 09

Aim : To provision cloud infrastructure using Terraform.

LO Mapped : LO4

LO : Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

- **IAM User / Account ID:** Experiment7 / 702561770822
- **Target AWS Region:** ap-southeast-2 (Asia Pacific - Sydney)
- **Deployment Status:** Successfully Provisioned, Verified, and Deprovisioned

1. Aim / Objective

To design, write, validate, and execute Infrastructure as Code (IaC) configurations using HashiCorp Terraform to dynamically provision, configure, verify, and destroy an Amazon Elastic Compute Cloud (EC2) virtual machine and associated resources on Amazon Web Services (AWS).

2. Theoretical Background

What is Infrastructure as Code (IaC)?

Infrastructure as Code (IaC) is an IT infrastructure provisioning process where developers and DevOps engineers automatically manage and provision cloud resources (networks, virtual machines, load balancers, and storage) through human-readable definition files rather than manual user interface configurations. IaC enforces version control, repeatability, scalability, and consistency across environments.

Key Terraform Concepts & Lifecycle Commands

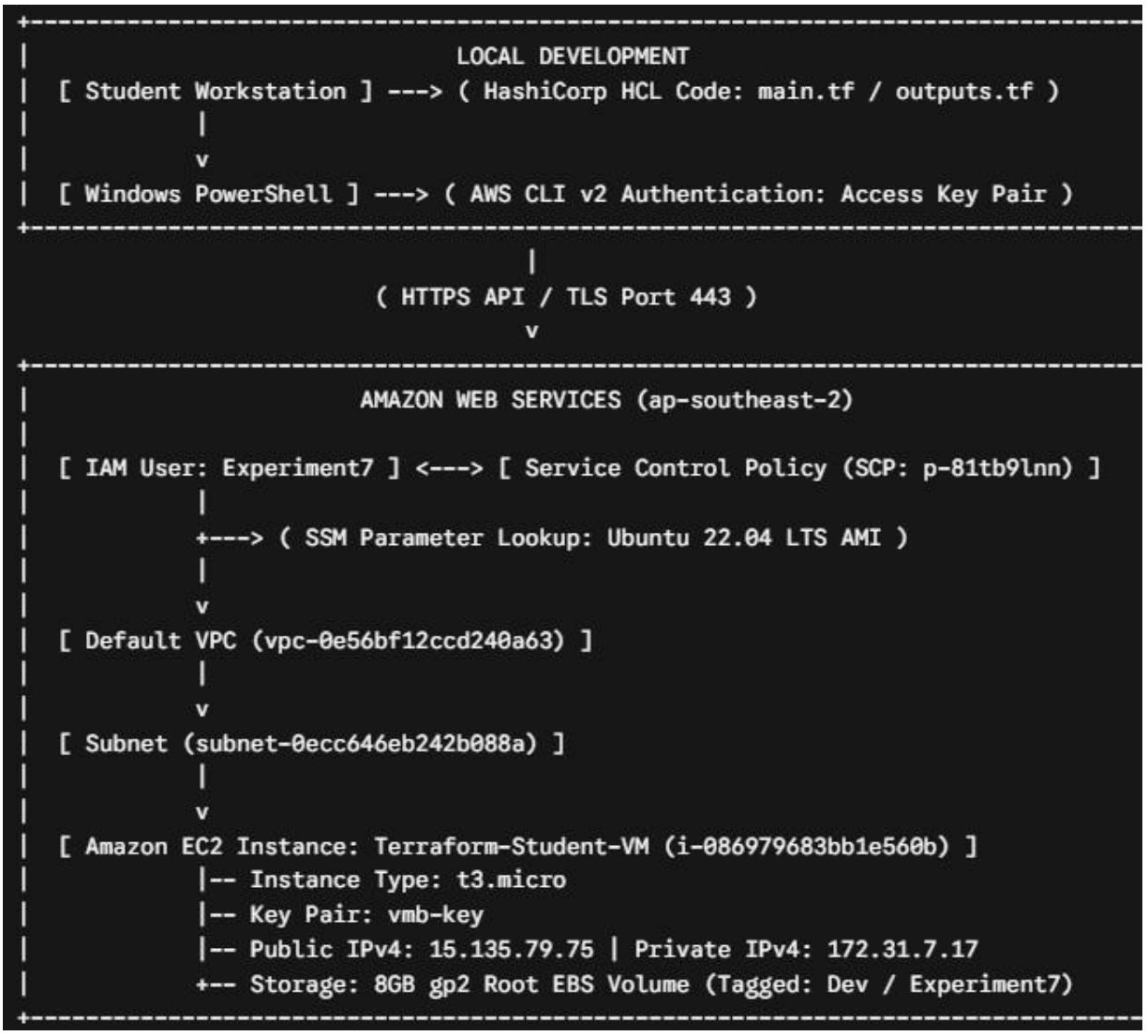
HashiCorp Terraform is an open-source IaC tool that uses HashiCorp Configuration Language (HCL) to define cloud infrastructure declarative state files.

- **Declarative Configuration (.tf):** Files where target cloud resources are defined. All .tf files in a single execution directory are parsed together as a unified configuration.
- **terraform init:** Initializes the working directory, downloads required provider plugins (e.g., hashicorp/aws v6.67.0), and creates dependency lock files (.terraform.lock.hcl).
- **terraform validate:** Checks the syntax, structure, and internal consistency of HCL configuration files.
- **terraform plan:** Compares the local declarative configuration against the active cloud infrastructure state and constructs an execution plan without applying changes.
- **terraform apply:** Executes the proposed plan against AWS APIs to provision or modify cloud resources, saving state metadata into terraform.tfstate.
- **terraform output:** Extracts and displays defined output variables from the state file.
- **terraform destroy:** Deprovisions and cleans up all managed resources from AWS to prevent unnecessary billing.

Interfacing AWS Concepts & Security

1. **AWS Identity and Access Management (IAM):** Grants precise execution policies (AmazonEC2FullAccess, AmazonVPCFullAccess, AmazonSSMReadOnlyAccess) and security credentials (Access Key ID & Secret Access Key) for programmatic CLI access.
2. **AWS Systems Manager (SSM) Parameter Store:** Dynamically retrieves system parameters such as up-to-date, region-specific Canonical Ubuntu AMI IDs (/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id).
3. **AWS Service Control Policies (SCPs):** Guardrails enforced at the AWS Organization level (p-81tb9lnn) that mandate resource compliance, such as explicit resource/volume tags (Environment = "Dev", Project = "Experiment7"), allowed instance types (t3.micro), and attached SSH Key Pairs (vmb-key).

3. Architecture Diagram & Workflow



4. Declarative Configuration Code

The configuration code was organized inside C:\Terraform.

1. main.tf (Infrastructure Definition)

Terraform

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
```

```
}

provider "aws" {
  region = "ap-southeast-2"
}

# Dynamic SSM Parameter Lookup for Ubuntu 22.04 LTS AMI
data "aws_ssm_parameter" "ubuntu_ami" {
  name = "/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id"
}

# Default VPC and Subnet Fetching
data "aws_vpc" "default" {
  default = true
}

data "aws_subnets" "default" {
  filter {
    name   = "vpc-id"
    values = [data.aws_vpc.default.id]
  }
}

# EC2 Instance Resource Definition
resource "aws_instance" "student_vm" {
  ami           = data.aws_ssm_parameter.ubuntu_ami.value
```

```
  instance_type = "t3.micro"
  subnet_id     = data.aws_subnets.default.ids[0]
  key_name      = "vmb-key"

  tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }

  volume_tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
}
```

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}

provider "aws" {
  region = "ap-southeast-2"
}

data "aws_ssm_parameter" "ubuntu_ami" {
  name = "/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id"
}

data "aws_vpc" "default" {
  default = true
}

data "aws_subnets" "default" {
  filter {
    name   = "vpc-id"
    values = [data.aws_vpc.default.id]
  }
}

resource "aws_instance" "student_vm" {
  ami           = data.aws_ssm_parameter.ubuntu_ami.value
  instance_type = "t3.micro"
  subnet_id     = data.aws_subnets.default.ids[0]
  key_name      = "vmb-key"

  tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
```

```
  volume_tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
}
```

2. outputs.tf (Output State Extraction)

Terraform

output "instance_id" {

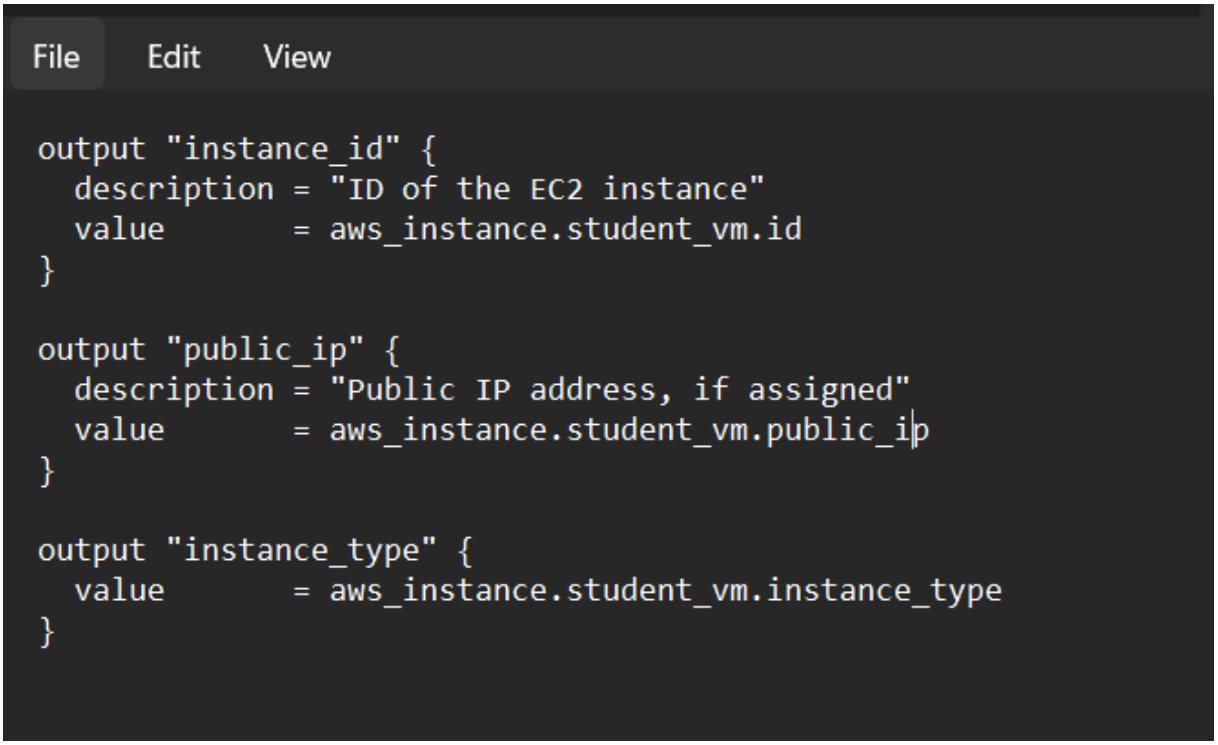
description = "ID of the created EC2 instance"

value = aws_instance.student_vm.id

}

```
output "instance_type" {
  description = "Instance type of the created VM"
  value       = aws_instance.student_vm.instance_type
}

output "public_ip" {
  description = "Public IP address assigned to the EC2 instance"
  value       = aws_instance.student_vm.public_ip
}
```



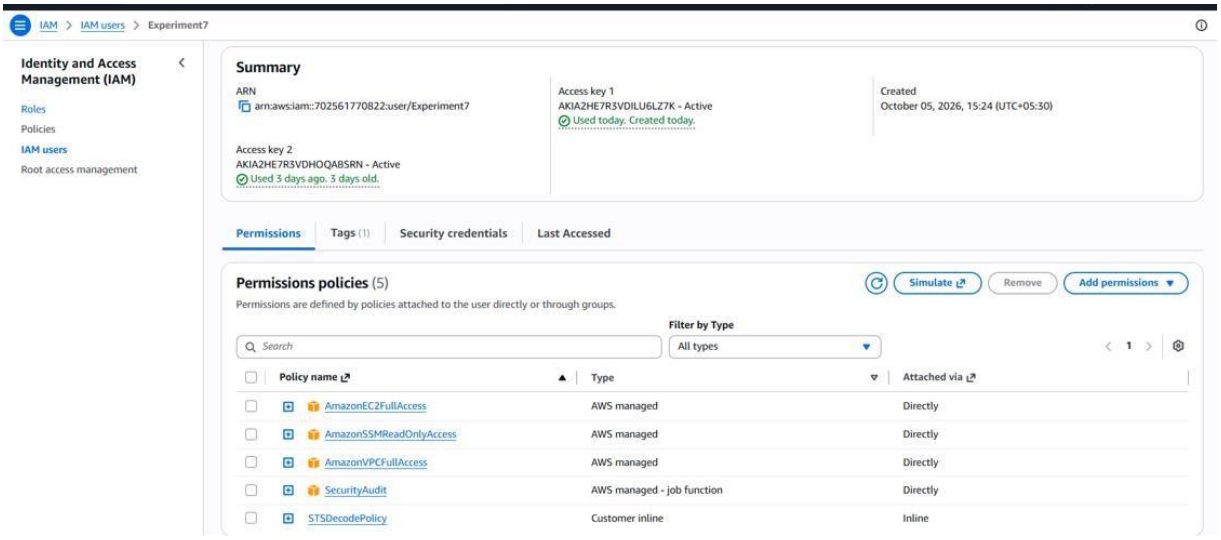
5. Step-by-Step Implementation Procedure

Phase 1: Local Environment & Credentials Configuration

Step 1: AWS IAM Credentials & Access Key Setup

1. Log in to the AWS Management Console and navigate to **IAM** > **Users** > **Experiment7**.
2. Select **Security credentials**, click **Create access key**, and choose **Command Line Interface (CLI)**.
3. Generate and record the credentials:

- **Access Key ID:** AKIA2HE7R3VDILU6LZ7K
- **Secret Access Key:** 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



Step 2: AWS CLI Authentication & Verification

1. Open Windows PowerShell and run aws configure:

PowerShell

AWS Access Key ID: AKIA2HE7R3VDILU6LZ7K

AWS Secret Access Key: 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

Default region name: ap-southeast-2

Default output format: json

2. Verify identity via STS:

PowerShell

aws sts get-caller-identity

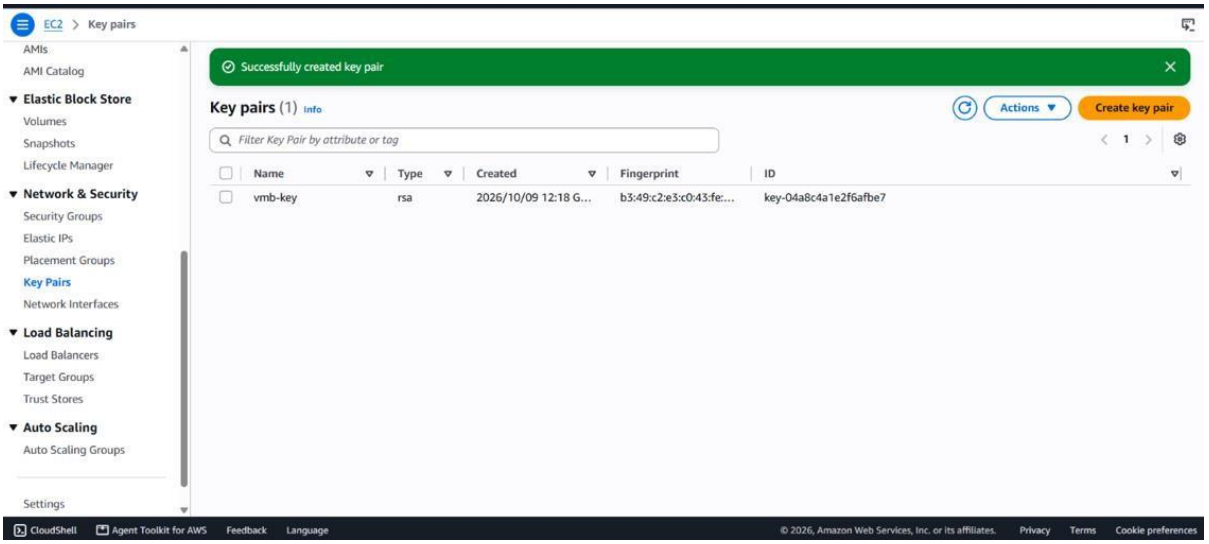
.

Step 3: EC2 Key Pair Generation in ap-southeast-2

1. Navigate to **AWS Console > EC2 > Key Pairs** in region ap-southeast-2.
2. Click **Create key pair**:
   - **Name:** vmb-key
   - **Key pair type:** RSA

- **Private key file format:** .pem

3. Download and save vmb-key.pem.

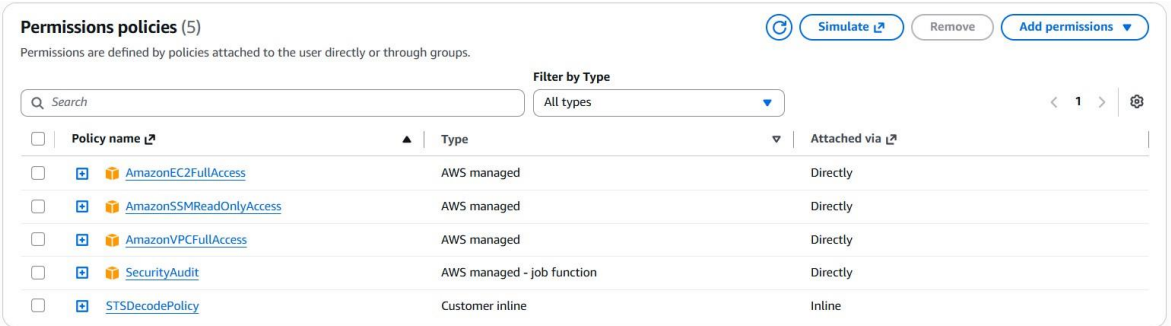


Phase 2: Resolving IAM Permissions & Policy Extensions

Step 4: Attaching SSM Read-Only Permissions

To enable Terraform to query the Canonical Ubuntu AMI ID dynamically from AWS Parameter Store, attach the SSM policy:

1. Go to **IAM > Users > Experiment7 > Permissions**.
2. Click **Add permissions > Attach policies directly**.
3. Search and select **AmazonSSMReadOnlyAccess**.
4. Click **Add permissions**.



Phase 3: Terraform Execution Lifecycle

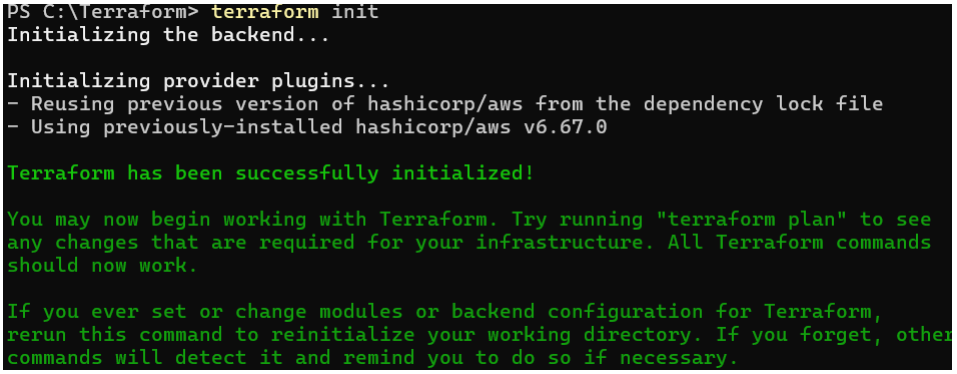
Step 5: Directory Setup & Terraform Initialization

1. Create directory C:\Terraform and save main.tf and outputs.tf inside.
2. Open PowerShell, navigate to directory, and initialize backend:

PowerShell

cd C:\Terraform

terraform init



Step 6: Code Validation & Execution Planning

1. Validate syntax:

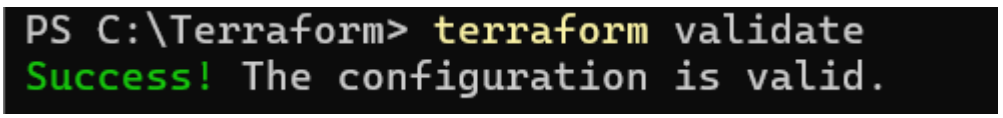
PowerShell

terraform validate

2. Generate execution plan:

PowerShell

terraform plan



```
Windows PowerShell

PS C:\Terraform> terraform plan
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 2s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 3s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 0s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

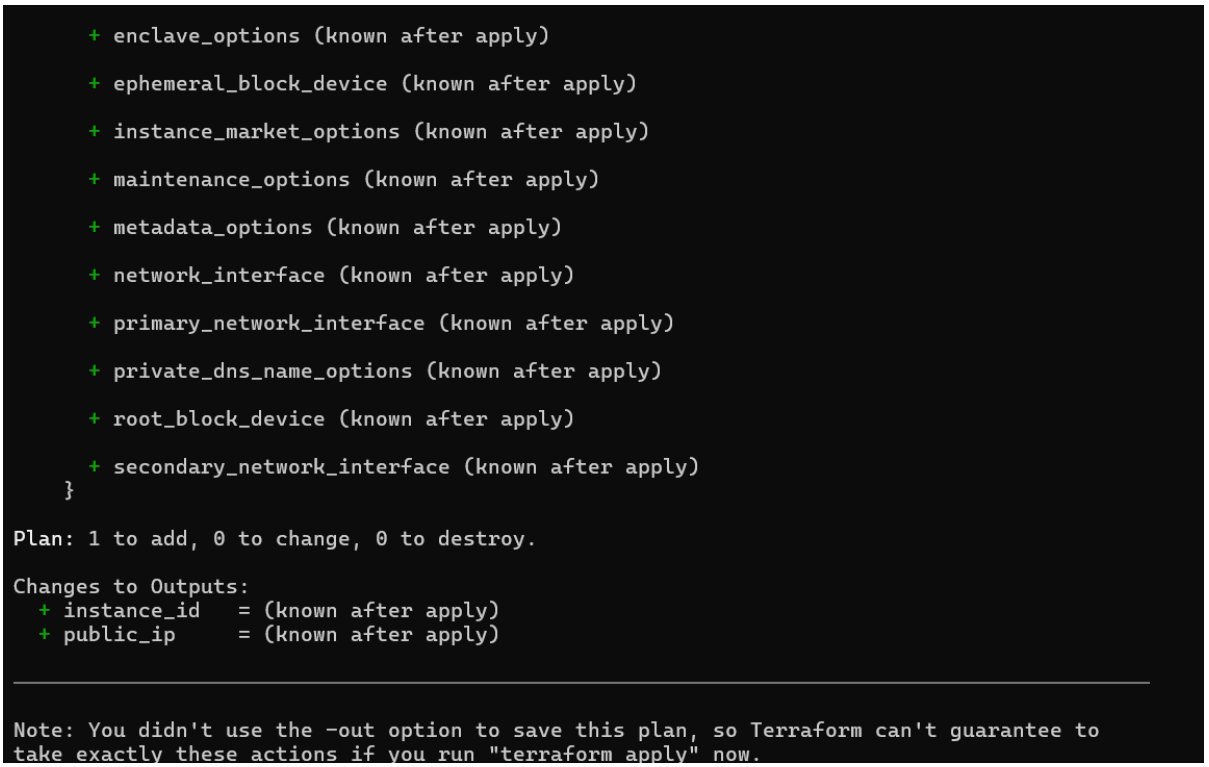
  # aws_instance.student_vm will be created
  + resource "aws_instance" "student_vm" {
      + ami                                  = (sensitive value)
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "vmb-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
```

```
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-southeast-2"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = "subnet-0ecc646eb242b088a"
      + tags                                 = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tags_all                             = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + volume_tags                          = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)
```



Step 7: Applying Infrastructure Provisioning

1. Execute apply command:

PowerShell

terraform apply

2. Enter **yes** at the approval prompt.
3. Confirm resource creation completion.

```
PS C:\Terraform> terraform apply
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 1s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 2s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 1s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

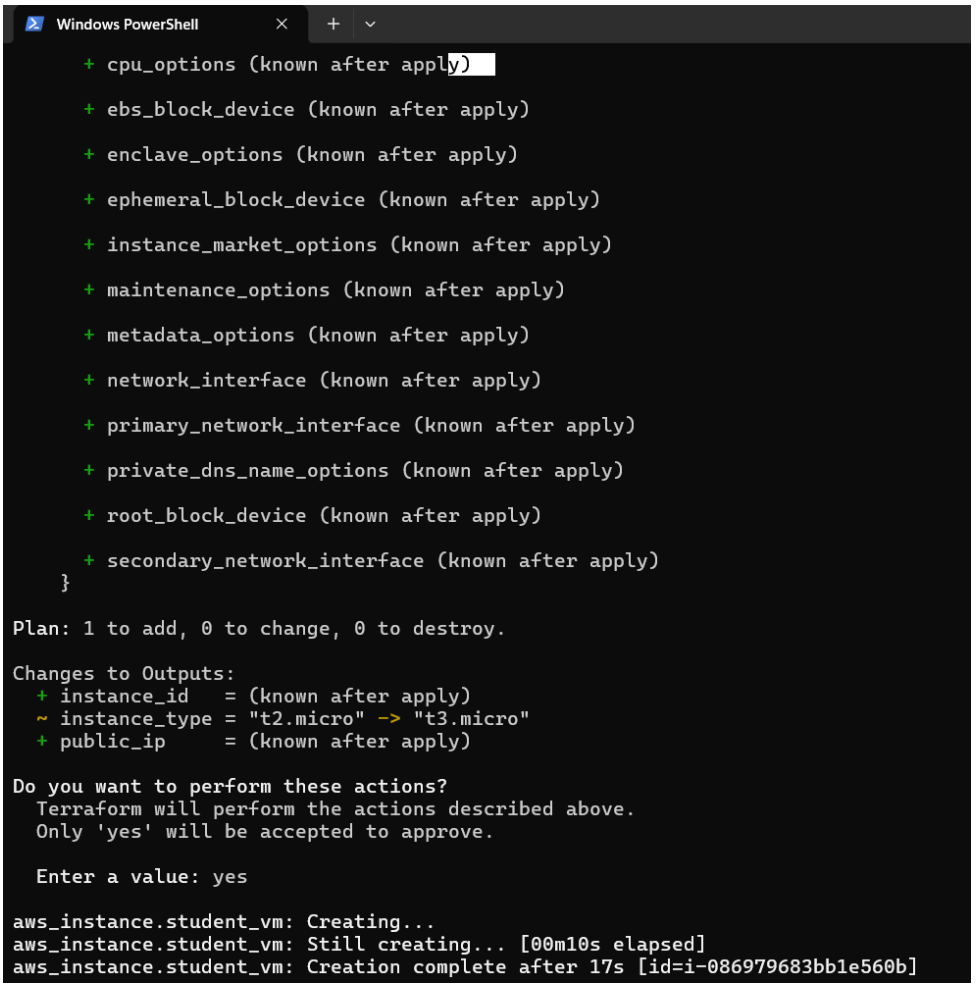
  # aws_instance.student_vm will be created
  + resource "aws_instance" "student_vm" {
      + ami                                  = (sensitive value)
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "vmb-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
```

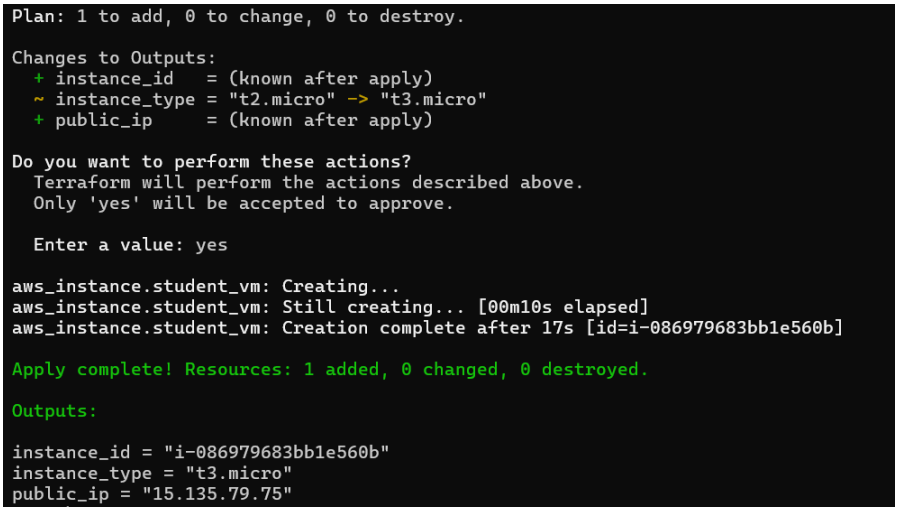
Windows PowerShell

```
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-southeast-2"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = "subnet-0ecc646eb242b088a"
      + tags                                 = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tags_all                             = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + volume_tags                          = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)
```





Phase 4: Verification & Teardown

Step 8: Verifying Terraform Outputs & AWS EC2 Console State

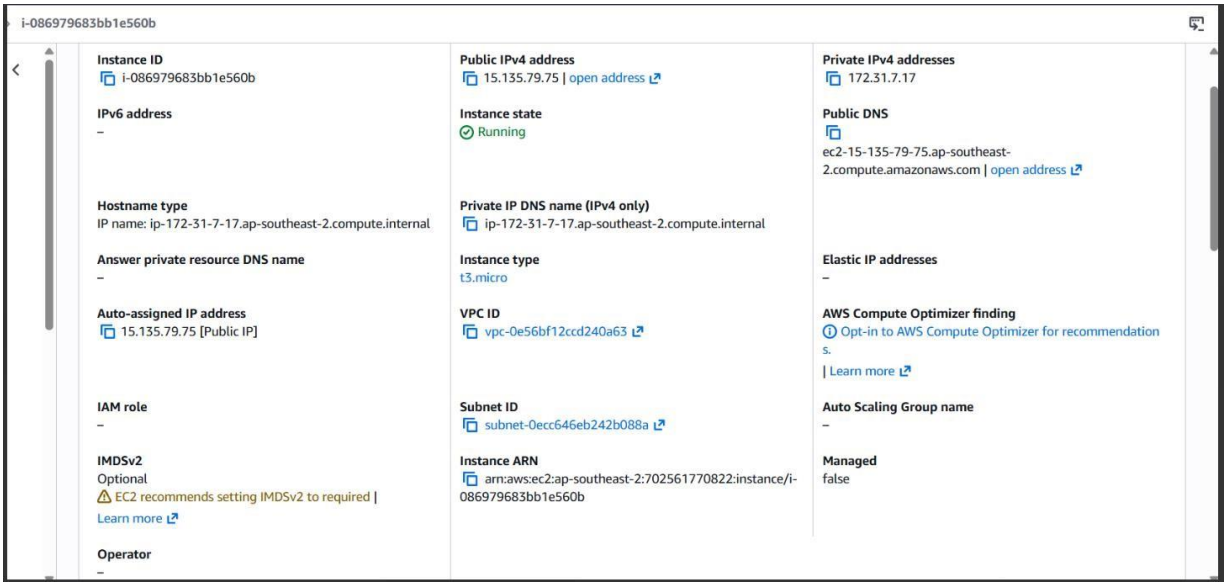
1. Run output command in PowerShell:

PowerShell

terraform output

2. Open browser and navigate to **AWS EC2 Dashboard** > **Instances** in region ap-southeast-2.
3. Verify that Terraform-Student-VM (i-086979683bb1e560b) is running with public IP 15.135.79.75.

```
PS C:\Terraform> terraform output
instance_id = "i-086979683bb1e560b"
instance_type = "t3.micro"
public_ip = "15.135.79.75"
```



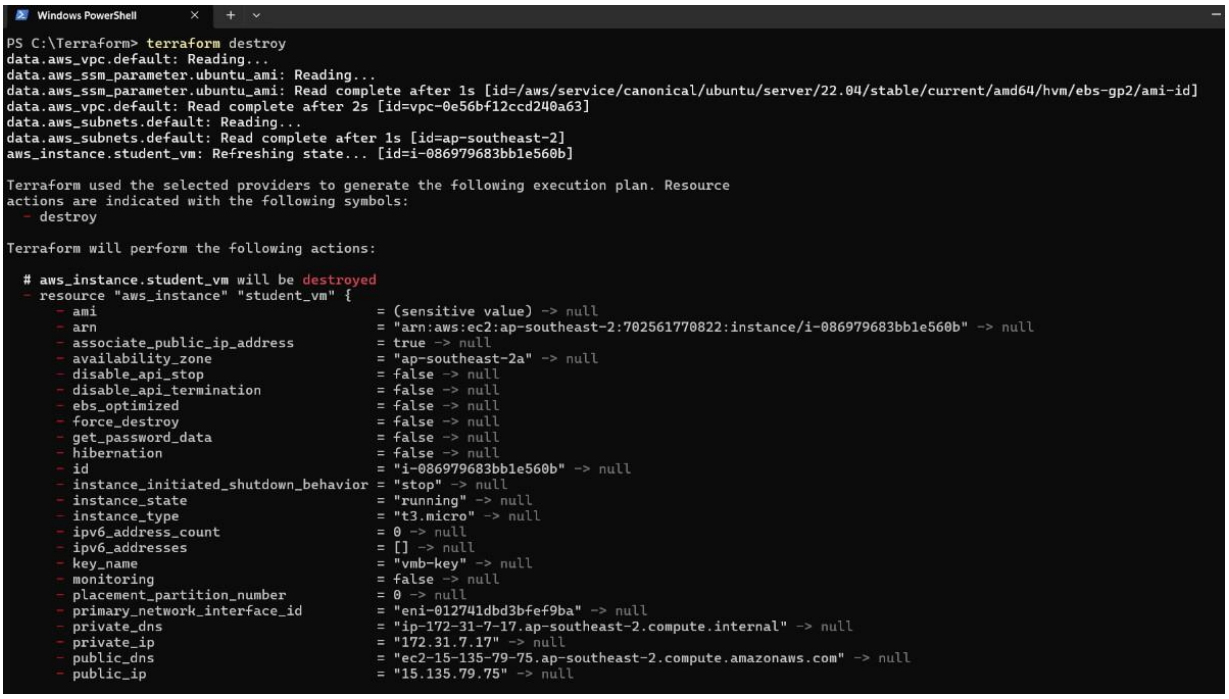
Step 9: Deprovisioning Cloud Resources (terraform destroy)

1. In PowerShell, initiate teardown:

PowerShell

terraform destroy

2. Type **yes** when prompted.
3. Confirm resource destruction.



```
Windows PowerShell

PS C:\Terraform> terraform destroy
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 1s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 2s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 1s [id=ap-southeast-2]
aws_instance.student_vm: Refreshing state... [id=i-086979683bb1e560b]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_instance.student_vm will be destroyed
  - resource "aws_instance" "student_vm" {
      - ami                                  = (sensitive value) -> null
      - arn                                  = "arn:aws:ec2:ap-southeast-2:702561770822:instance/i-086979683bb1e560b" -> null
      - associate_public_ip_address          = true -> null
      - availability_zone                    = "ap-southeast-2a" -> null
      - disable_api_stop                     = false -> null
      - disable_api_termination              = false -> null
      - ebs_optimized                        = false -> null
      - force_destroy                        = false -> null
      - get_password_data                    = false -> null
      - hibernation                          = false -> null
      - id                                   = "i-086979683bb1e560b" -> null
      - instance_initiated_shutdown_behavior = "stop" -> null
      - instance_state                       = "running" -> null
      - instance_type                        = "t3.micro" -> null
      - ipv6_address_count                   = 0 -> null
      - ipv6_addresses                       = [] -> null
      - key_name                             = "vmb-key" -> null
      - monitoring                           = false -> null
      - placement_partition_number           = 0 -> null
      - primary_network_interface_id         = "eni-012741dbd3bfef9ba" -> null
      - private_dns                          = "ip-172-31-7-17.ap-southeast-2.compute.internal" -> null
      - private_ip                           = "172.31.7.17" -> null
      - public_dns                           = "ec2-15-135-79-75.ap-southeast-2.compute.amazonaws.com" -> null
      - public_ip                            = "15.135.79.75" -> null
```



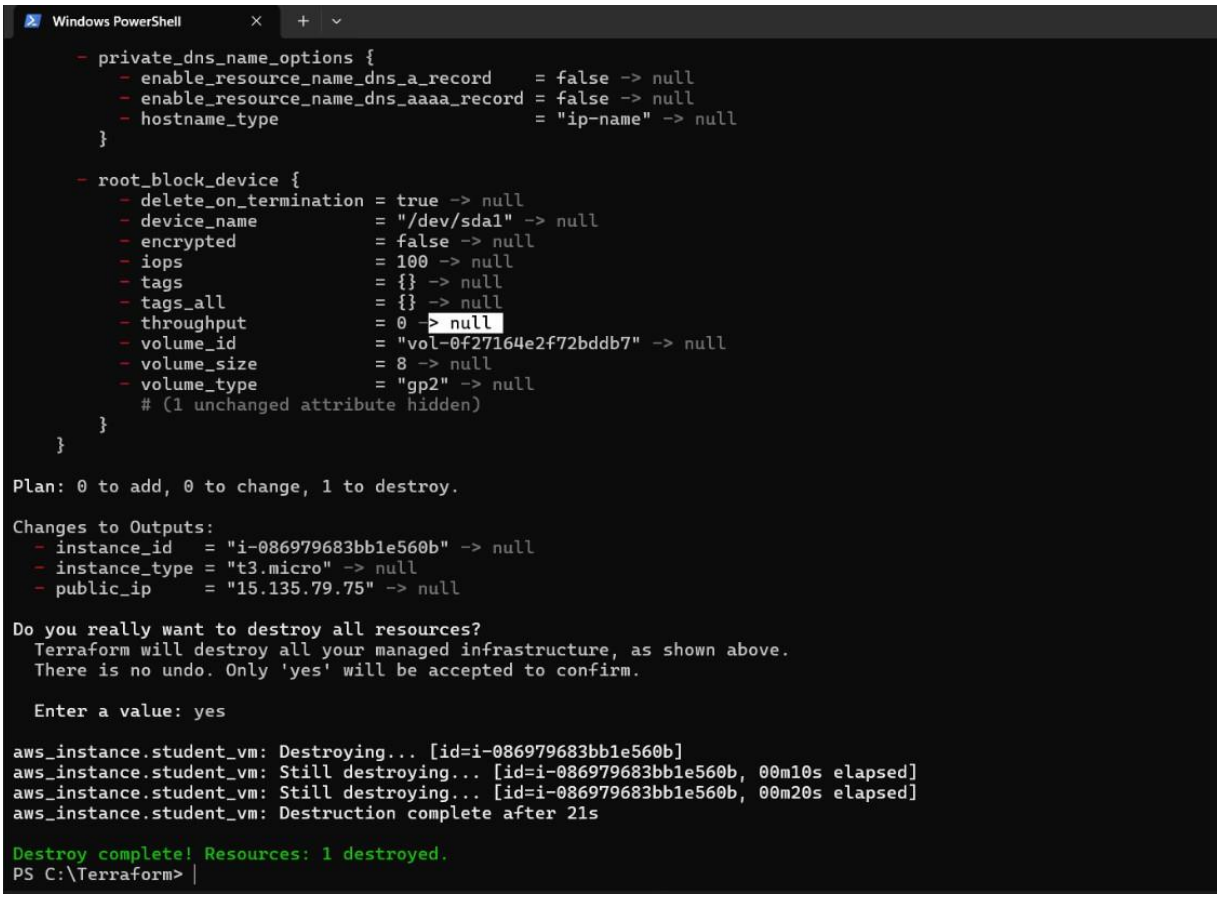
```
Windows PowerShell

      - public_dns                           = "ec2-15-135-79-75.ap-southeast-2.compute.amazonaws.com" -> null
      - public_ip                            = "15.135.79.75" -> null
      - region                               = "ap-southeast-2" -> null
      - secondary_private_ips                = [] -> null
      - security_groups                      = [
          - "default",
        ] -> null
      - source_dest_check                    = true -> null
      - subnet_id                            = "subnet-0ecc646eb242b088a" -> null
      - tags                                 = {
          - "Environment" = "Dev"
          - "Name"        = "Terraform-Student-VM"
          - "Project"     = "Experiment7"
        } -> null
      - tags_all                             = {
          - "Environment" = "Dev"
          - "Name"        = "Terraform-Student-VM"
          - "Project"     = "Experiment7"
        } -> null
      - tenancy                              = "default" -> null
      - user_data_replace_on_change          = false -> null
      - volume_tags                          = {
          - "Environment" = "Dev"
          - "Name"        = "Terraform-Student-VM"
          - "Project"     = "Experiment7"
        } -> null
      - vpc_security_group_ids               = [
          - "sg-0b3493b246c88ea0a",
        ] -> null
        # (8 unchanged attributes hidden)

      - capacity_reservation_specification {
          - capacity_reservation_preference = "open" -> null
        }

      - cpu_options {
          - core_count            = 1 -> null
          - threads_per_core      = 2 -> null
            # (2 unchanged attributes hidden)
        }
```


Windows PowerShell
}
- credit_specification {
- cpu_credits = "unlimited" -> null
}
- enclave_options {
- enabled = false -> null
}
- maintenance_options {
- auto_recovery = "default" -> null
}
- metadata_options {
- http_endpoint = "enabled" -> null
- http_protocol_ipv6 = "disabled" -> null
- http_put_response_hop_limit = 1 -> null
- http_tokens = "optional" -> null
- instance_metadata_tags = "disabled" -> null
}
- primary_network_interface {
- delete_on_termination = true -> null
- network_interface_id = "eni-012741dbd3bfef9ba" -> null
}
- private_dns_name_options {
- enable_resource_name_dns_a_record = false -> null
- enable_resource_name_dns_aaaa_record = false -> null
- hostname_type = "ip-name" -> null
}
- root_block_device {
- delete_on_termination = true -> null
- device_name = "/dev/sda1" -> null
- encrypted = false -> null
- iops = 100 -> null
- tags = {} -> null
- tags_all = {} -> null
- throughput = 0 -> null



6. Real-World Technical Challenges & Solutions

Sr No.	Error / Issue Encountered	Root Cause	Engineering Solution Implemented
1	403 UnauthorizedOperation on ec2:DescribeImages	Organization Service Control Policy (SCP: p-81tb9lnn) blocked AMI search queries.	Replaced data "aws_ami" block with dynamic Systems Manager (SSM) Parameter Store lookup (aws_ssm_parameter).
2	AccessDeniedException on ssm:GetParameter	User Experiment7 lacked permissions to read SSM paths.	Attached managed IAM policy AmazonSSMReadOnlyAccess to IAM user Experiment7.
3	403 Explicit Deny on ec2:RunInstances	SCP enforced mandatory tagging on both EC2 instances and root EBS volumes.	Configured both tags and volume_tags blocks (Environment="Dev",

			Project="Experiment7") in HCL code.
4	InvalidParameterCombination: Not eligible for Free Tier	Sandbox environment enforced t3.micro instance type over t2.micro.	Updated instance_type = "t3.micro" in main.tf.
5	InvalidAMIID.Malformed	Hardcoded AMI ID mismatch across AWS regions.	Bound AMI lookup dynamically to active region (ap-southeast-2) via SSM Parameter Store.

7. Summary of Provisioned Infrastructure Attributes

Parameter Name	Deployed Value	Source / Verification Method
AWS Account ID	702561770822	aws sts get-caller-identity
IAM User	Experiment7	AWS IAM Console / CLI
AWS Region	ap-southeast-2 (Sydney)	HCL Provider Configuration
Instance Name Tag	Terraform-Student-VM	main.tf HCL Block
Instance ID	i-086979683bb1e560b	terraform output / AWS Console
Instance Type	t3.micro	AWS EC2 Instance Details
Public IPv4 Address	15.135.79.75	terraform output / AWS Console
Private IPv4 Address	172.31.7.17	AWS EC2 Instance Details
VPC ID	vpc-0e56bf12ccd240a63	AWS VPC Console / data.aws_vpc
Subnet ID	subnet-0ecc646eb242b088a	AWS Subnet Console / data.aws_subnets

Key Pair	vmb-key	AWS EC2 Key Pairs / main.tf
Lifecycle Final State	Terminated	terraform destroy / AWS Console

8. Conclusion

Learning Outcomes Achieved

Through this experiment, hands-on mastery was acquired in defining, provisioning, managing, and deprovisioning AWS cloud infrastructure using HashiCorp Terraform as an Infrastructure as Code (IaC) tool. The experiment successfully satisfied **Lab Outcome 4 (LO4)** by executing automated resource management workflows on cloud infrastructure.

Applications of the Studied Technique in Industry

- **Automated & Repeatable Deployments:** Eliminates manual human error in cloud console management by declaring infrastructure state in version-controlled repositories.
- **Multi-Cloud Architecture:** Terraform provides a unified framework to provision resources across AWS, Azure, GCP, and Kubernetes clusters using declarative HCL scripts.
- **Policy Compliance & Governance:** Seamlessly integrates with enterprise Service Control Policies (SCPs) and security frameworks to enforce automated tagging, network isolation, and access controls.

Engineering Relevance

Infrastructure as Code forms the foundational backbone of modern DevOps paradigms, Site Reliability Engineering (SRE), and CI/CD pipelines. Programmatically controlling infrastructure lifecycle allows engineering teams to rapidly create ephemeral testing

environments, scale compute capacity dynamically, maintain exact environment parity across staging and production, and implement disaster recovery protocols.